Hackathon Finder

Workflow Document

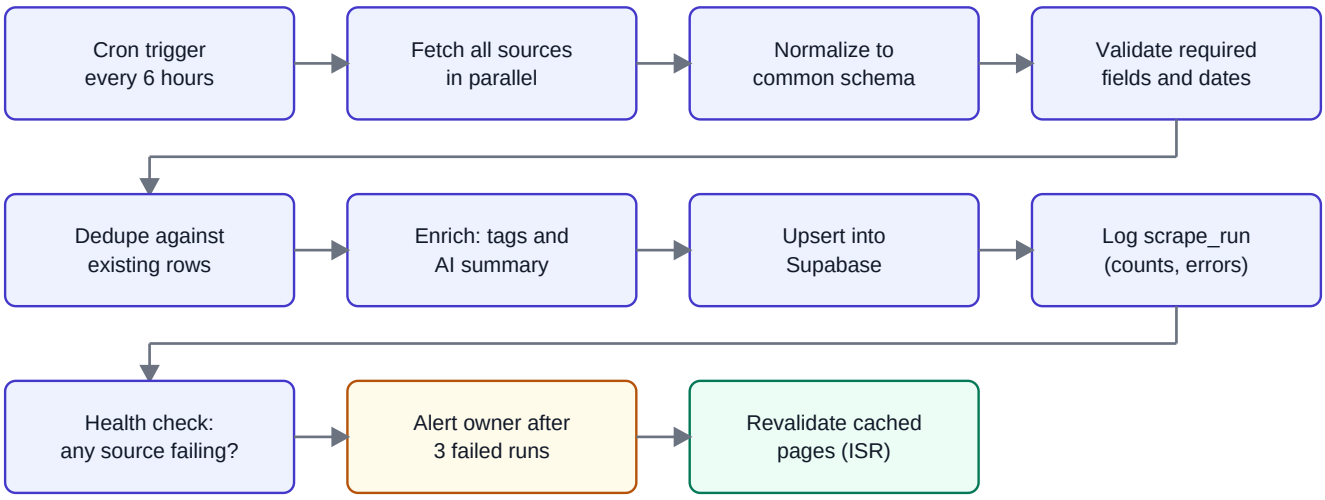
Version 1.0 | 14 Oct 2026 | How data, users and developers move through the system

Overview

Five workflows make up the product: data ingestion, user discovery, alerts, submission and moderation, and the developer release flow. Each is shown below as a step diagram followed by notes on edge cases.

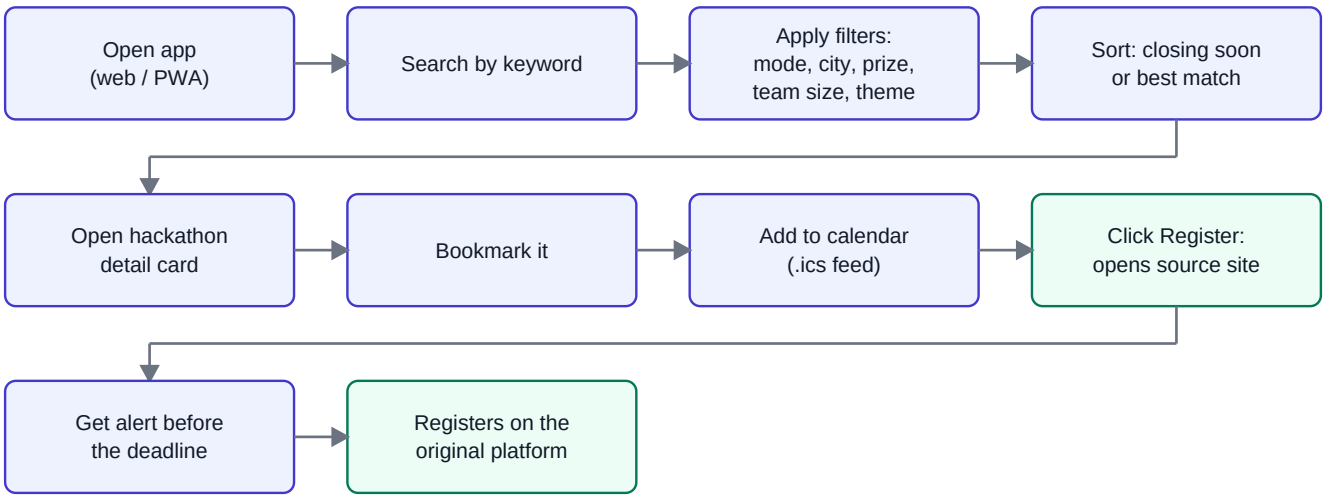
Workflow	Trigger	Actor	Output
Ingestion	Cron every 6 hours	GitHub Actions	Fresh, deduplicated listings in the database
Discovery	User opens the app	Student	Shortlist, bookmark, register on source site
Alerts	Cron every hour	GitHub Actions	Telegram and email messages
Submission	User submits the form	Organiser and admin	Approved listing goes live
Release	Git push	Developer	Preview and production deploys

1. Data ingestion workflow



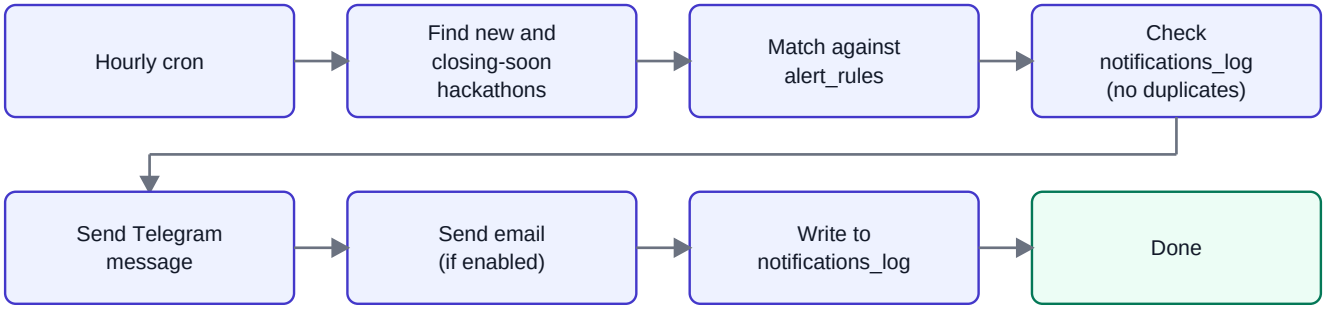
- A failed source is retried once, then skipped; the other sources still complete.
- Rows missing a title, link or any date are dropped and counted in the log.
- Enrichment runs only for new or changed rows, so AI usage stays inside the free tier.
- Listings not seen for 14 days are hidden, not deleted.

2. User discovery workflow



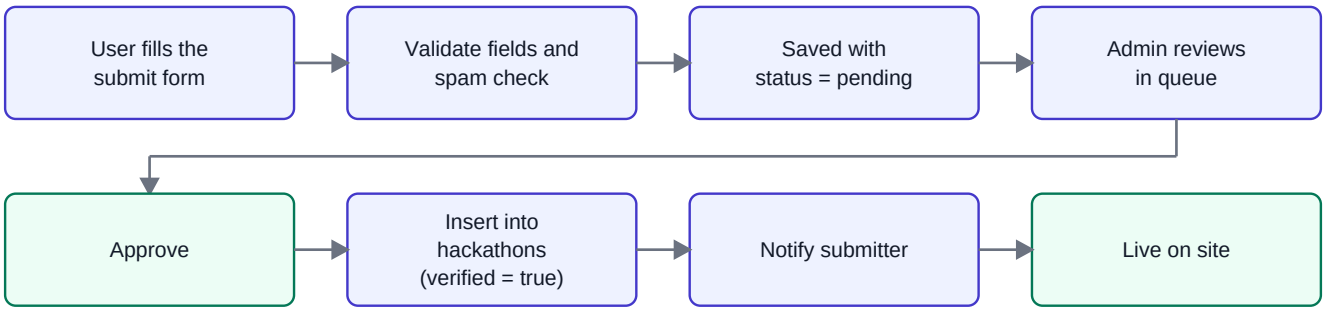
- Browsing needs no login; bookmarks, alerts and the calendar feed need an account.
- Closed events are hidden by default, with a toggle to show them.
- Each card always shows the source name and a direct link so users can verify details.

3. Alerts workflow



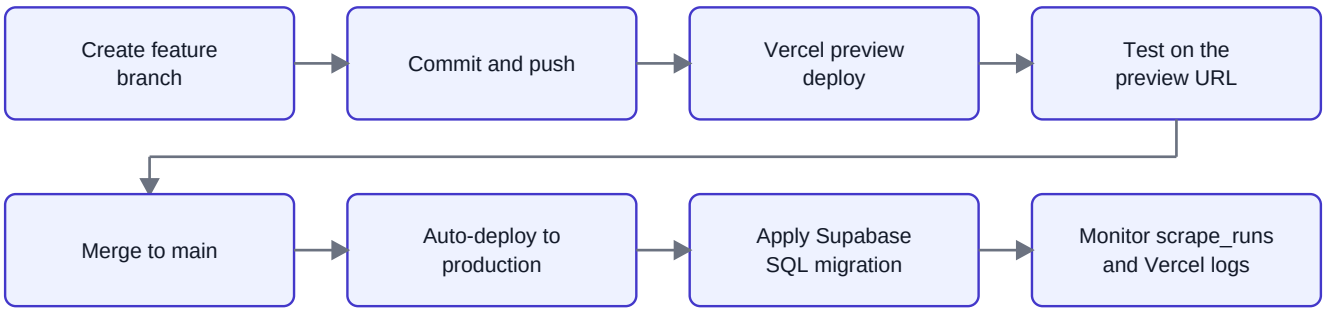
- Closing-soon alerts fire at 48 hours and 24 hours before the registration deadline.
- The unique key (user, hackathon, kind) in notifications_log stops repeat messages if a job reruns.
- A user can mute alerts from the Telegram bot with /mute.

4. Submission and moderation workflow



- Reject path: admin picks a reason, status becomes rejected, and the submitter is notified.
- The form is rate limited and has a hidden honeypot field to catch bots.
- Approved user-submitted events are labelled so people know the source.

5. Developer release workflow



Build timeline

Week	Focus	Done when
1	Schema, 2 scrapers, list page	Real hackathons from two sources show in the app
2	Search, filters, deadline radar, detail card	A student can find and open an event in under a minute
3	Accounts, bookmarks, calendar feed, submit and moderation	A submitted event can be approved and appear live
4	Alerts, AI summaries, polish, launch	Alerts arrive on Telegram; shared in campus groups